New bounds for small two- and four-colour Ramsey numbers

Lachlan Dauth Claude Opus 5.5

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Abstract

We give upper bounds for the Ramsey numbers $R(3,k)$ with $24 \le k \le 50$, for which Radziszowski's survey lists no upper bounds. They come from an exact bootstrap of lower bounds on the number of edges of triangle-free graphs with bounded independence number, starting from the published values and lower bounds of Goedgebeur and Radziszowski; for instance $R(3,24) \le 201$, $R(3,30) \le 297$ and $R(3,50) \le 729$. With in addition the table of Backelin's unrefereed report, the bounds are 1 to 3 lower for $24 \le k \le 36$ except $k = 35$. Deleting vertices from the quartic residue graphs of Lindsay and Cain gives $R(4,21) \ge 260$ and $R(4,24) \ge 370$. Finally, the four-colour form of the neighbourhood recurrence, with $R(3,4,4) \le 73$ and $R(4,4,4) \le 198$ from a companion paper and the survey's $R(3,3,3,4) \le 149$, gives $R(3,3,4,4) \le 442$, $R(3,4,4,4) \le 1521$ and $R(4,4,4,4) \le 6082$. All certificates and witnesses are checked exactly by programs in the ancillary files.

1 Introduction

For integers $s, t \ge 1$, the Ramsey number $R(s,t)$ is the least n such that every red-blue colouring of the edges of K_n contains a red K_s or a blue K_t; equivalently, every graph on n vertices contains a K_s or an independent set of size t. Multicolour numbers $R(t_1,\dots,t_r)$ are defined in the same way. The current state of knowledge is collected in Radziszowski's dynamic survey [9], whose revision 18 we use for all comparisons, together with the later preprint [12].

This paper is a companion to a study of three-colour numbers [3], in which local Ramsey bounds inside colour neighbourhoods are combined with double counting and with flag-algebra certificates. It gives new bounds for two-colour numbers, the upper bounds by adapting the ladder procedure of [3, Section 5] and the lower bounds by deleting vertices from quartic residue graphs, and derives consequences for four-colour numbers. Our results are listed in Table 1.

number	previous	new	where
$R(3,k)$, $24 \le k \le 50$	no upper bound [9]	Table 2	Theorem 2.1
$R(4,21)$	≥ 242 [11]; ≥ 257 (implicit in [12])	≥ 260	Theorem 3.1
$R(4,24)$	≥ 343 (implicit in [8])	≥ 370	Theorem 3.1
$R(3,3,4,4)$	≤ 450 [9]	≤ 442	Theorem 4.1
$R(3,4,4,4)$	≤ 1576 [9] (implicit in [2])	≤ 1521	Theorem 4.1
$R(4,4,4,4)$	≤ 6306 [9]; ≤ 6301 (implicit in [2])	≤ 6082	Theorem 4.1

Table 1: Our results. For $R(3,k)$ with $k \ge 24$, iterating $R(3,k) \le R(3,k-1) + k$, which is strict when both terms are even, or Shearer's bound [10], gives weaker explicit bounds (Table 2). Where the two rows of Table 2 differ, the smaller bound also uses the unrefereed report [1].

The upper bounds for $R(3,k)$ (Section 2) and the lower bounds for $R(4,k)$ (Section 3) are computer-assisted; floating-point solvers and local search were used only to find certificates and

witnesses, and every certificate and witness is checked in exact arithmetic by a program that does not share code with the search (Section 5). The four-colour bounds (Section 4) follow by hand from $\mathrm{R}(3,3,3,4) \le 149$ [9] and from $\mathrm{R}(3,4,4) \le 73$ and $\mathrm{R}(4,4,4) \le 198$, which are proved in [3, Theorems 8.3 and 8.2] by certificates checked in exact arithmetic by computer.

2 Upper bounds for $\mathrm{R}(3,k)$

A $(3,k;n)$-graph is a triangle-free graph on n vertices with no independent set of size k, a $(3,k;n,e)$-graph is one with e edges, and $e(3,k,n)$ is the least such e. As in [3, Section 3], $m_k(n)$ is a *valid lower bound* if every $(3,k;n)$-graph has at least $m_k(n)$ edges. For a vertex v of a $(3,k;n,e)$-graph G, $N(v)$ is independent, so $d_v \le k-1$, and $G_v = G - N[v]$ is a $(3,k-1;n-1-d_v)$-graph, as an independent set of G_v extends by v. As no edge lies inside $N(v)$, G_v has $e - Z_v$ edges, where $Z_v = \sum_{u \in N(v)} d_u$. As $\sum_v Z_v = \sum_v d_v^2$, valid lower bounds m_{k-1} give

$$\sum_v \left(d_v^2 + m_{k-1}(n-1-d_v)\right) \le ne,$$

the inequality of Graver and Yackel [6] on which Goedgebeur and Radziszowski [5] base their bounds for $k \ge 11$; it is the two-colour analogue of the density inequality [3, Lemma 3.3]. As $ne = \sum_v nd_v/2$, it says $\sum_v \varphi(d_v) \le 0$, where $\varphi(d) = d^2 + m_{k-1}(n-1-d) - nd/2$.

We bootstrap as in the ladder of [3, Section 5], with this inequality in place of the density inequality. Level $k \ge 11$ uses the bound B_{k-1} on $\mathrm{R}(3,k-1)$ and the valid lower bounds m_{k-1} from level $k-1$; level 11 starts from $B_{10} = 42$ [5, Theorem 6] and the values and lower bounds for $e(3,10,n)$ in [5]. Level k treats the orders $n = 0, 1, 2, \ldots$ in turn, records a valid lower bound $m_k(n)$, and stops at the first order it excludes, B_k; so $\mathrm{R}(3,k) \le B_k$. At order n the admissible degrees are the $d \le \min(k-1, n-1)$ with $n - 1 - d \le B_{k-1} - 1$. The order n is excluded if $\varphi(d) > 0$ at every admissible d, or if every possible e, up to $\lfloor n(k-1)/2 \rfloor$, is excluded by the step below. Otherwise $m_k(n)$ is the larger of the bound of [5, Theorem 2] and $\lceil n\alpha/(2\beta) \rceil$, for a line $\alpha - \beta d$ with $\beta > 0$ through an edge of the lower convex hull of the points $(d, \varphi(d))$, d admissible: the line lies below φ at every admissible d, so $0 \ge \sum_v \varphi(d_v) \ge n\alpha - 2\beta e$. Deleting a vertex as in [3, Lemma 3.5(a)], which gives $m_k(n-1) + \max\{0, n - B_{k-1}\}$, never does better.

We add one step. When these bounds give $m_k(n) = e_0$, we enumerate every degree sequence with sum $2e_0$, admissible entries and largest entry at most $e_0 - m_k(n-1)$ (as $G - v$ is a $(3,k;n-1)$-graph) that satisfies the inequality with $e = e_0$. We then try to exclude each one by a small linear programme in the numbers of edges inside and between degree classes. Its constraints hold for every $(3,k;n,e_0)$-graph with that degree sequence:

- the edge ends at the class of degree a number a times the class size;
- the sum of Z_v over the class of degree a is at most the class size times $e_0 - m_{k-1}(n-1-a)$;
- it is at least the class size times $e_0 - \lfloor (n-1-a)(k-2)/2 \rfloor$, as G_v has maximum degree at most $k-2$;
- the number of edges between two classes is at most the product of their sizes;
- Mantel's bound holds on every union of classes;
- a vertex of degree a has at least t neighbours of degree at most b, so the class of degree a sends at least t times its size edge ends to classes of degree at most b. Here t is least such that the t smallest entries at most b, other than that of the vertex itself, together with $a - t$ times the least entry above b, sum to at most $e_0 - m_{k-1}(n-1-a)$; this sum is a lower bound for Z_v when v has exactly t such neighbours, and it decreases as t grows.

If every sequence is excluded, so is e_0, and the step is repeated with $e_0 + 1$. Each exclusion is certified either by the failure of the neighbour-count test, when for some class no such t exists, or by rational Farkas multipliers for named constraints.

Theorem 2.1 (computer-assisted)**.** *From the published values and lower bounds of [5, Theorem 2 and Tables 3–5] and* $\mathrm{R}(3,10) \le 42$ *[5, Theorem 6],* $\mathrm{R}(3,k)$ *is at most the value in the row "[5] only" of Table 2 for* $24 \le k \le 50$*. With in addition the lower bounds on* $e(3,k,n)$ *in the table of [1, Section 5], it is at most the value in the row "with [1]".*

k	24	25	26	27	28	29	30	31	32
[5] only	201	216	231	247	264	280	297	315	333
with [1]	200	215	230	245	261	278	295	312	331
k	33	34	35	36	37	38	39	40	41
[5] only	351	370	389	409	429	450	471	492	513
with [1]	349	369	389	408	429	450	471	492	513
k	42	43	44	45	46	47	48	49	50
both	536	559	582	606	630	654	679	704	729

Table 2: Upper bounds for $\mathrm{R}(3,k)$ from Theorem 2.1. The two rows coincide for $k = 35$ and $k \ge 37$, and for $k \ge 42$ they are shown as one, "both". The survey [9] lists no upper bounds for these k. For comparison, $\mathrm{R}(3,k) \le \mathrm{R}(3,k-1) + k$, strict when both terms are even, iterated from the survey's $\mathrm{R}(3,23) \le 185$, gives 209, 234, 259, 371 and 1171 for $k = 24, 25, 26, 30, 50$. Shearer's bound [10] $\mathrm{R}(3,k) \le \lfloor (k-1)/f(k-1) \rfloor + 1$, with $f(d) = (d \ln d - d + 1)/(d-1)^2$, gives 223 for $k = 24$ and 792 for $k = 50$.

Above level 10, each chain uses only the bounds B_{k-1} that it proves itself, never the unpublished values of [9], and neither needs the bound $\mathrm{R}(3,10) \le 41$ listed in [9]. The report [1] is a preprint that gives little of its proofs, so the second row rests on inputs without published proofs. The first row is 1 to 3 larger for $24 \le k \le 34$ and $k = 36$, and equals the second for $k = 35$ and $k \ge 37$. Without the step above, the chain from [5] alone gives the same bounds for $11 \le k \le 50$, except $\mathrm{R}(3,20) \le 146$ and $\mathrm{R}(3,35) \le 390$. The chain from [5] alone also raises seven lower bounds of [5, Tables 9–11] by one, to $e(3,13,58) \ge 261$, $e(3,14,69) \ge 366$, $e(3,14,70) \ge 382$ and $e(3,15,n) \ge 516$, 534, 553 and 573 for $n = 82, \ldots, 85$; Backelin's unpublished tables may already contain these.

The certificates are checked by a program that does not import the generating code: it enumerates the degree sequences itself, rebuilds each named constraint from its definition, and verifies the multipliers in exact arithmetic. The chain with [1] has one row for each level $11 \le k \le 50$ and each order $0 \le n < B_k$, 13,376 rows in all. It excludes 765 degree sequences at 53 triples (k,n,e), 686 by the neighbour-count test and 79 by Farkas multipliers. The chain from [5] alone has 13,403 rows and excludes 766 sequences at 50 triples, 684 and 82 respectively.

For $k \le 23$ the procedure improves no value of [9]. From [5] alone it gives $\mathrm{R}(3,17) \le 109$ and $\mathrm{R}(3,20) \le 145$, which [9] credits to an unpublished manuscript of Backelin and to Lesser's thesis [7], respectively. Adding the two values $e(3,11,39) \ge 119$ and $e(3,12,43) \ge 129$ of [1] lowers its $\mathrm{R}(3,16) \le 98$ and $\mathrm{R}(3,19) \le 133$ to 97 and 132, as stated in [1]. With the whole table of [1] it gives $\mathrm{R}(3,23) \le 185$, against 187 from [5] alone. The survey [9] credits this bound to a personal communication of Backelin; it now has a derivation that can be checked, given the inputs of [1]. At $k = 18$, 21 and 22 this chain gives 121, 158 and 172, one more than the unpublished values of [9] (from [5] alone, 121, 159 and 173). For $k = 18$, at $n = 120$ the hull already gives $e \ge 1020 = 120 \cdot 17/2$, so only the 17-regular sequence remains, and in it every G_v would be a $(3,17;102,731)$-graph. So $e(3,17,102) \ge 732$, one more than our bound, would give $\mathrm{R}(3,18) \le 120$.

3 Lower bounds for $\mathrm{R}(4,k)$

For a prime $p \equiv 1 \pmod 8$ let Q_p be the circulant graph on $\mathbb{Z}_p$ in which $x \sim y$ if and only if $x - y$ is a nonzero fourth power; the relation is symmetric, as -1 is a fourth power. Lindsay and Cain [8] showed that Q_{313} and Q_{457} are K_4-free with no independent sets of size 22 and 25, respectively; their independence numbers are $\alpha(Q_{313}) = 21$ and $\alpha(Q_{457}) = 24$. If a vertex set S meets every independent set of size $\alpha(Q_p)$, then $Q_p - S$ is K_4-free and has independence number less than $\alpha(Q_p)$.

Theorem 3.1 (witnesses checked by computer)**.** $\mathrm{R}(4,21) \geq 260$ *and* $\mathrm{R}(4,24) \geq 370$.

Proof. The maps $x \mapsto ax + b$ with a a nonzero fourth power are automorphisms of Q_p. Let O_1, O_2, O_3 be the cosets of the group of nonzero fourth powers in $\mathbb{Z}_p^*$ other than the group itself, in a fixed order, and let r_j be the least element of O_j. Let $k \geq 2$, let J be an independent k-set and let $u \in J$. The translate $J - u = \{x - u : x \in J\}$ is an independent k-set that contains 0, so its other elements are not fourth powers and lie in $O_1 \cup O_2 \cup O_3$. Let j be least with $J - u$ meeting O_j, say at w. Then r_j/w is a fourth power, and multiplying by it maps $J - u$ to an independent k-set that contains 0 and r_j and still avoids $O_1, \ldots, O_{j-1}$, as multiplication by a fourth power fixes each coset. Call such a set a representative. So every independent k-set is the image of a representative under one of the maps above. An exhaustive search lists all representatives for $(p,k) = (313,21)$ and $(457,24)$. A weighted local search found sets S of 54 and 88 vertices, listed in the ancillary files, and an exact program checks that each meets every image of every representative. So $Q_{313} - S$ is a K_4-free graph on 259 vertices with no independent 21-set, and $Q_{457} - S$ one on 369 vertices with no independent 24-set. □

Verification. Both graphs are rebuilt from the lists of deleted vertices by programs written independently of the search. The 259-vertex graph is also checked by a direct exhaustive search for K_4 and for independent 21-sets, and the 369-vertex graph by one for K_4 and for independent 24-sets. Neither search uses symmetry, so each is independent of the orbit argument above. □

The previous bounds are as follows. For $\mathrm{R}(4,21)$, the survey has 242 [11], and $\mathrm{R}(4,20) \geq 252$ from a recent preprint [12] gives 257 by [9, 2.3(e)]. For $\mathrm{R}(4,24)$, the non-neighbourhood of a vertex of Q_{457} (Q_{457} is 114-regular) has 342 vertices, is K_4-free and has no independent 24-set, as such a set would extend by the vertex. This gives $\mathrm{R}(4,24) \geq 343$. The survey lists no value for $\mathrm{R}(4,24)$; its rules give 324, by applying [9, 2.3(e)] twice to $\mathrm{R}(4,22) \geq 314$ [8]. The method does not reach $\mathrm{R}(4,23) \geq 320$, which would improve the bound 319 that [9, 2.3(e)] gives from $\mathrm{R}(4,22) \geq 314$: that needs a set of at most $457 - 319 = 138$ vertices meeting every independent 23-set of Q_{457}, and the smallest we found has 147.

4 Four colours

For $\mathbf{t} = (t_1, \ldots, t_4)$ and the unit vectors $\mathbf{e}_i$, the four-colour form of the neighbourhood recurrence [9, 6.1(a)] is $\mathrm{R}(\mathbf{t}) \leq \sum_{i=1}^{4} \mathrm{R}(\mathbf{t} - \mathbf{e}_i) - 2$. Indeed, in a colouring of K_n with no colour-j K_{t_j} for any j, the colour-i neighbourhood of a vertex v contains no colour-i $K_{t_i - 1}$, which would extend by v, and no colour-j K_{t_j} for $j \neq i$, so it has at most $\mathrm{R}(\mathbf{t} - \mathbf{e}_i) - 1$ vertices; summing over i gives $n - 1 \leq \sum_i (\mathrm{R}(\mathbf{t} - \mathbf{e}_i) - 1)$. The inequality is strict when its right-hand side is even and some term $\mathrm{R}(\mathbf{t} - \mathbf{e}_i)$ is even: at the order one less, which is odd, the count is tight, so every vertex has colour-i degree exactly $\mathrm{R}(\mathbf{t} - \mathbf{e}_i) - 1$, which is odd, contradicting the handshake lemma. Both statements hold with the same proofs when each $\mathrm{R}(\mathbf{t} - \mathbf{e}_i)$ is replaced by a proven upper bound for it. An entry equal to 2 may be dropped, leaving a number with fewer colours, as a colouring with no colour-i K_2 has no colour-i edges.

Theorem 4.1. $\mathrm{R}(3,3,4,4) \leq 442$, $\mathrm{R}(3,4,4,4) \leq 1521$ *and* $\mathrm{R}(4,4,4,4) \leq 6082$.

Proof. With $R(3,4,4) \le 73$ [3, Theorem 8.3] and $R(3,3,3,4) \le 149$ [9], the recurrence gives $R(3,3,4,4) \le 2 \cdot 73 + 2 \cdot 149 - 2 = 442$; parity does not apply, as every term is odd. With $R(4,4,4) \le 198$ [3, Theorem 8.2] it then gives $R(3,4,4,4) \le 198 + 3 \cdot 442 - 2 = 1522$, which is even, as are all four terms, so $R(3,4,4,4) \le 1521$. Finally $R(4,4,4,4) \le 4 \cdot 1521 - 2 = 6082$, and parity does not apply, as 1521 is odd. □

The two inputs from [3] are computer-assisted. The survey's $R(3,3,3,4) \le 149$ is itself the recurrence with parity, $3 \cdot 30 + 62 - 2 = 150$, from $R(3,3,4) = 30$ and $R(3,3,3,3) \le 62$ [4]. With the survey's $R(3,3,4,4) \le 450$ in place of 442, that is, without $R(3,4,4) \le 73$, the bound $R(4,4,4) \le 198$ alone gives $R(3,4,4,4) \le 1545$ and $R(4,4,4,4) \le 6178$. We have not tried the analogue of the parity clause modulo 3 of [2], which gives the survey's $R(3,4,4,4) \le 1576$.

We also tried to improve $R(3,3,3,3) \le 62$ [4], that is, to exclude colourings of K_{61} with four triangle-free colours, by certificates analogous to those for $R(4,4,4)$ in [3, Sections 7 and 8]. Such a certificate exists only if a flag relaxation, here with four- or five-vertex patterns and with four-colour analogues of the local caps of [3, Lemmas 2.1 and 2.3] among its constraints, is infeasible. At $n = 61$ every colour degree is at most $R(3,3,3) - 1 = 16$ and the four sum to 60, so the *profile* of a vertex, its sorted quadruple of colour degrees, is $(12,16,16,16)$, $(13,15,16,16)$, $(14,14,16,16)$, $(14,15,15,16)$ or $(15,15,15,15)$. Both relaxations are numerically feasible: floating-point solvers found an assignment of pattern densities meeting all their constraints, even when every vertex is required to have profile $(12,16,16,16)$. The four-vertex relaxation was solved as a linear programme by HiGHS, with its semidefinite constraint imposed by eigenvector cuts until no eigenvalue was below -10^{-8}, and the five-vertex relaxation as a semidefinite programme by Clarabel at tolerance 10^{-8}. So we obtained no such certificate from these relaxations, and no bound follows.

5 Verification, literature and limitations

Ancillary files. The ancillary directory, the same as for [3], contains the programs and data: `two_colour_r3k/` for Theorem 2.1, with the inputs from [5] and [1] and the certificates of both chains, and `two_colour_r4k/` for Theorem 3.1, with the lists of deleted vertices, the adjacency matrices of the two graphs and the logs of the direct searches. The two inputs of Theorem 4.1 from [3] are checked by the programs of that paper in the same directory. The checkers are written in Python and C with their standard libraries; the C programs use POSIX threads. Negative controls (a weakened input, a removed multiplier or sequence certificate, a value raised by one, a forged cap, a deleted vertex put back) are all rejected.

Literature. We searched arXiv and the web up to 23 October 2026 for earlier statements of our new bounds, and found none. For $R(3,k)$ with $k \ge 24$ the survey lists no upper bounds, and the upper bounds of [5] and of Lesser's thesis [7] stop at $k = 16$ and $k = 21$; Backelin's unpublished manuscript, to which [9] credits $R(3,17) \le 109$, may contain bounds of this kind. For $R(4,21)$ the best previous bound we know of, 257, follows from $R(4,20) \ge 252$ in the preprint [12], which we have not checked; without it the bound is 242 [11]. For $R(4,24)$ the survey lists nothing, and 343 is implicit in [8]. For the four-colour numbers the previous bounds are those of [9] and [2] in Table 1. Such a search cannot establish priority, and we would welcome pointers to any earlier source.

Limitations. For $k \le 23$ the bootstrap of Section 2 improves none of the bounds in [9]; at $k = 18$ it would give $R(3,18) \le 120$ if its own lower bound $e(3,17,102) \ge 731$ were one larger. The second row of Table 2 rests on the lower bounds of [1], whose proofs are largely unpublished. The deletion method of Section 3 does not reach $R(4,23) \ge 320$. The four-colour bounds rest on $R(3,4,4) \le 73$ and $R(4,4,4) \le 198$ [3, Theorems 8.3 and 8.2], which are computer-assisted,

and on $\mathrm{R}(3,3,3,4) \le 149$, which [9, 6.1(a)] derives from $\mathrm{R}(3,3,4) = 30$ and $\mathrm{R}(3,3,3,3) \le 62$ [4], both computer-assisted.

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